Honors Geometry Practice Test 4

Problems 116–168

Name: _____

Show all work. Give all answers in exact, simplified form with rationalized denominators unless told otherwise.

Part 1: Essential Skills

Skill 1: Vectors between points

Rule: The vector from A to B is $\overrightarrow{AB} = B - A$ (subtract coordinates). The length of $[a, b]$ is $\sqrt{a^2 + b^2}$.

Example: From $A = (2, -1)$ to $B = (5, 3)$: $\overrightarrow{AB} = [5 - 2,\ 3 - (-1)] = [3, 4]$, and its length is $\sqrt{9 + 16} = 5$.

a. Find the vector from $(-4, 6)$ to $(2, -2)$, and find its length.

b. Find a vector with integer components, neither of them zero, that has length $\sqrt{68}$.

c. Is there a vector with integer components that has length $\sqrt{21}$? Explain.

Skill 2: Rationalizing denominators

Rule: Multiply the top and bottom by the same thing, so the value doesn't change. For a single square root, multiply by that root. For a sum or difference, multiply by the **conjugate**, which flips the sign: $(a+\sqrt{b})(a-\sqrt{b}) = a^2 - b$.

Examples: $\dfrac{6}{\sqrt{3}} \cdot \dfrac{\sqrt{3}}{\sqrt{3}} = \dfrac{6\sqrt{3}}{3} = 2\sqrt{3}$ $\qquad$ $\dfrac{8}{3+\sqrt{5}} \cdot \dfrac{3-\sqrt{5}}{3-\sqrt{5}} = \dfrac{8(3-\sqrt{5})}{9-5} = 6 - 2\sqrt{5}$

Rationalize each denominator and simplify.

a. $\dfrac{10}{\sqrt{5}}$

b. $\dfrac{4}{\sqrt{7}-\sqrt{3}}$

c. $\dfrac{\sqrt{3}}{\sqrt{3}+1}$

Skill 3: Scaling a vector

Rule: To get length 1, divide each component by the length. To get length k, multiply the length-1 vector by k. For the opposite direction, multiply by $-k$.

Example: $[3, 4]$ has length 5. Same direction, length 1: $\left[\frac{3}{5}, \frac{4}{5}\right]$. Opposite direction, length 15: $-15\left[\frac{3}{5}, \frac{4}{5}\right] = [-9, -12]$.

a. Given $[5, -12]$, find the vector with length 1 in the same direction, and the vector with length 26 in the opposite direction.

b. Given $[3, -1]$, find the vector with length 1 in the same direction, and the vector with length 5 in the opposite direction.

Skill 4: Perpendicular vectors

Rule: Swap the components and change one sign: $[a, b]$ is perpendicular to both $[-b, a]$ and $[b, -a]$. (Check with slopes: $\frac{b}{a}$ and $-\frac{a}{b}$ are opposite reciprocals.)

Example: $[3, 4]$ is perpendicular to $[-4, 3]$ and $[4, -3]$.

a. Find both vectors that are perpendicular to $[7, 1]$ and have length $5\sqrt{2}$.

b. Find both vectors that are perpendicular to $[-8, 6]$ and have length 20.

Skill 5: A point part of the way along a segment

Rule: The point that is k of the way from A to B is $A + k\,\overrightarrow{AB}$. The midpoint is the case $k = \frac{1}{2}$.

Example: One-third of the way from $A = (1, 2)$ to $B = (7, 11)$: $\overrightarrow{AB} = [6, 9]$, so the point is $(1, 2) + \frac{1}{3}[6, 9] = (3, 5)$.

a. Find the point that is three-fourths of the way from $(-5, 6)$ to $(7, -2)$.

b. Point $P = (4, 3)$ is two-thirds of the way from $A = (-4, 9)$ to B. Find B.

Skill 6: Adding vectors

Rule: Add vectors component by component. Following one vector and then another (head-to-tail) is the same as following their sum. Translating a point by a vector adds the vector to the point.

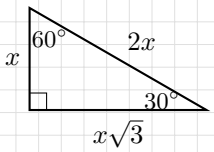
Example: $[2, 5] + [5, 7] = [7, 12]$. Translating $(3, 8)$ by $[2, -5]$ gives $(5, 3)$.

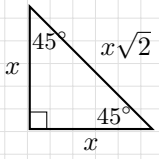
a. Starting at point S, a bug moves with a displacement vector of $[5, -2]$ and then with another displacement vector $[1, 6]$. What single vector brings it straight back to S?

b. Find numbers s and t so that $s\,[2, 1] + t\,[1, 3] = [7, 11]$.

Skill 7: Special right triangles

Ratios: In a 30-60-90 triangle, the sides are x (opposite 30°), $x\sqrt{3}$ (opposite 60°), and $2x$ (hypotenuse). In a 45-45-90 triangle, the sides are x, x, and $x\sqrt{2}$.





Example (30-60-90): If the long leg is 6, then $x\sqrt{3} = 6$, so $x = \dfrac{6}{\sqrt{3}} = 2\sqrt{3}$, and the hypotenuse is $2x = 4\sqrt{3}$.

Example (45-45-90): If the hypotenuse is 10, then $x\sqrt{2} = 10$, so $x = \dfrac{10}{\sqrt{2}} \cdot \dfrac{\sqrt{2}}{\sqrt{2}} = 5\sqrt{2}$. Each leg is $5\sqrt{2}$.

a. Fill in the missing sides.

Triangle	Short leg	Long leg	Hypotenuse
30-60-90	9		
30-60-90		$6\sqrt{3}$	
30-60-90		12	
30-60-90			14

Triangle	Leg	Leg	Hypotenuse
45-45-90	7		
45-45-90			18

b. An equilateral triangle has sides of length 10. How long is its altitude?

Skill 8: Perimeter of a special right triangle

Rule: Write each side in terms of x, add, factor out x, then solve and rationalize.

Example: A 45-45-90 triangle has perimeter 4. Then $x + x + x\sqrt{2} = 4$, so $x(2 + \sqrt{2}) = 4$, and $x = \dfrac{4}{2+\sqrt{2}} \cdot \dfrac{2-\sqrt{2}}{2-\sqrt{2}} = \dfrac{4(2-\sqrt{2})}{2} = 4 - 2\sqrt{2}$.

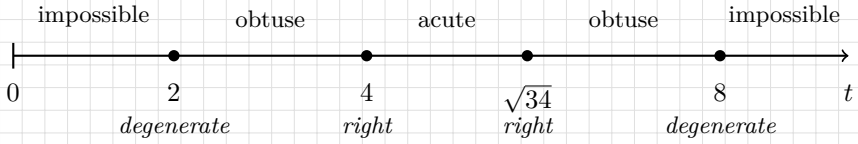
a. A 30-60-90 triangle has perimeter 18. Find the lengths of all three sides.

b. A 45-45-90 triangle has perimeter 10. Find the lengths of all three sides.

Skill 9: Triangle type from side lengths

Rule: A triangle has two fixed, **different** sides and a third side t. As t grows from 0, the triangle goes through: impossible → degenerate → obtuse → right → acute → right → obtuse → degenerate → impossible. A **degenerate** triangle has collapsed into a segment.

Example: sides 3, 5, and t. Degenerate: $t + 3 = 5$, so $t = 2$; and $3 + 5 = t$, so $t = 8$. Right with 5 longest: $t^2 + 3^2 = 5^2$, so $t = 4$. Right with t longest: $3^2 + 5^2 = t^2$, so $t = \sqrt{34}$.



a. A triangle has sides 8, 15, and t. Find each boundary value, then make a diagram like the example, labeling every boundary and every interval.



b. A triangle has sides 6, 6, and t. Make a similar diagram. Which parts of the pattern disappear? Explain why.



c. Give three side lengths that make an acute triangle, and three side lengths that make an obtuse triangle. Justify each set, including why it makes a triangle at all.

Skill 10: Reflections

Rules: Over the x-axis: $(x, y) \to (x, -y)$. Over the y-axis: $(x, y) \to (-x, y)$. Over $y = x$: $(x, y) \to (y, x)$. Over $y = -x$: $(x, y) \to (-y, -x)$.

Key idea: The mirror line is the perpendicular bisector of any point and its image.

Example: $P = (1, 2)$ reflects to $P' = (5, 4)$. The midpoint is $(3, 3)$, and $\overrightarrow{PP'} = [4, 2]$ has slope $\frac{1}{2}$, so the mirror line has slope -2, and the mirror line is $y - 3 = -2(x - 3)$.

a. Triangle ABC has $A = (1, 2)$, $B = (4, 3)$, and $C = (2, 5)$. Find the vertices of its image after a reflection over the line $y = -x$.

b. A reflection sends $(-3, 4)$ to $(5, 0)$. Find an equation of the mirror line.

Skill 11: Identifying a transformation

Rule: Read the vertices of a figure in order and check whether they go clockwise or counterclockwise. Translations and rotations **keep** the order; reflections and glide-reflections **reverse** it.

- **Translation:** every vertex moves by the same vector.
- **Rotation:** the figure turns about a fixed point (the center). The order is kept, but the vertices don't all move by the same vector.
- **Reflection:** order reversed, and one line is the perpendicular bisector of every point and its image.
- **Glide-reflection:** order reversed, but no single mirror line works.

Triangle ABC has $A = (1, 1)$, $B = (4, 1)$, and $C = (1, 3)$. Each triangle below is its image, listed with the images of A, B, and C in that order. Name each transformation and justify your answer. For a translation, give the vector; for a reflection, give the mirror line.

i. $(4, 5)$, $(7, 5)$, $(4, 7)$

ii. $(9, 1)$, $(6, 1)$, $(9, 3)$

iii. $(-1, 1)$, $(-1, 4)$, $(-3, 1)$

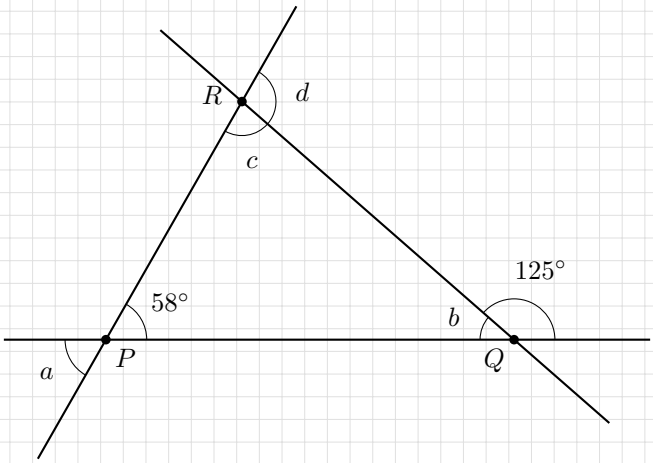
iv. $(5, -1)$, $(8, -1)$, $(5, -3)$

Skill 12: Angle hunts

Rules: Vertical angles (opposite each other where two lines cross) are equal. Angles that form a straight line (a linear pair) add to 180°. The three angles of a triangle add to 180°.

Example: Two lines cross, and one of the four angles is 40°. The angle vertical to it is also 40°, and each of the other two angles is $180° - 40° = 140°$.

Three lines intersect to form triangle PQR. The diagram is not drawn to scale.



a. Find a, b, c, and d. Give a reason for each.

b. Compare d with the angles of the triangle at P and Q. What do you notice? Explain why this happens.

Part 2: Putting Skills Together

Question 1

Let $A = (-1, 2)$ and $B = (3, 4)$. Point C is chosen so that triangle ABC is equilateral.

a. Find the side length of the triangle.

b. Find the length of the altitude from C to side $\overline{AB}$.

c. Find both possible locations of C.

Question 2

Let $A = (1, 2)$ and $B = (4, 8)$. Find the point that is exactly 5 units from A in the direction of B.

Question 3

Points $A = (0, 0)$ and $B = (8, 0)$ are fixed. Point C moves along the line $y = 4$, so at time t its position is $C = (0, 4) + t\,[1, 0] = (t, 4)$.

a. For which values of t is triangle ABC a right triangle? At which vertex is the right angle in each case?

b. For which values of t is the triangle acute? For which is it obtuse?

Question 4

Triangle ABC has $m\angle B = 45°$ and $m\angle C = 60°$. The altitude from A to side $\overline{BC}$ has length $6\sqrt{3}$. Find the lengths of all three sides and the perimeter of the triangle.

Question 5

A circular table is pushed into the corner of a room so that it touches both walls. A mark at point P on the edge of the table, in the curved gap between the table and the corner, is 2 inches from one wall and 9 inches from the other. Let r be the radius of the table.

a. Draw a diagram that includes the center of the table, point P, and a right triangle whose sides you can write in terms of r.

b. Write an equation using the Pythagorean Theorem. Expand, combine like terms, and factor to solve for r.

c. Your equation has two solutions. Which one is the radius of the table? Explain why the other one doesn't work.

Question 6

Triangle KLM has $K = (-4, 2)$, $L = (-3, 9)$, and $M = (-6, 8)$. Triangle XZY has $X = (0, 0)$, $Z = (5, 5)$, and $Y = (6, 2)$.

a. Show that $\triangle KLM \cong \triangle XZY$.

b. Explain why the transformation from KLM to XZY could be a reflection but cannot be a rotation or translation.

c. Find the equation of the mirror line. Then check that it works for all three pairs of vertices.

Question 7

a. Reflect $P = (2, 5)$ over the line $x = 1$, and then reflect the image over the line $x = 4$. Do the same for $Q = (-3, 0)$. What single transformation has the same effect? What if the reflections are done in the opposite order?

b. A point is translated by $[-5, 4]$ and then reflected over the y-axis. Its final image is $(1, 6)$. Where did the point start? What sequence of transformations would undo the original sequence?